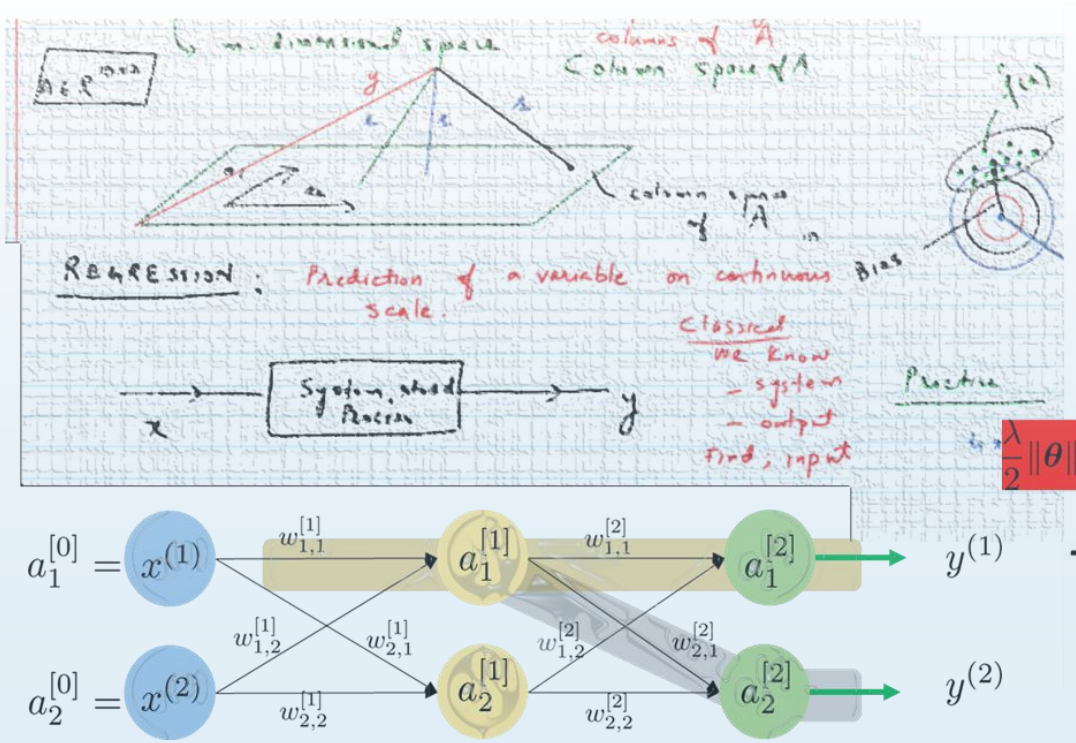
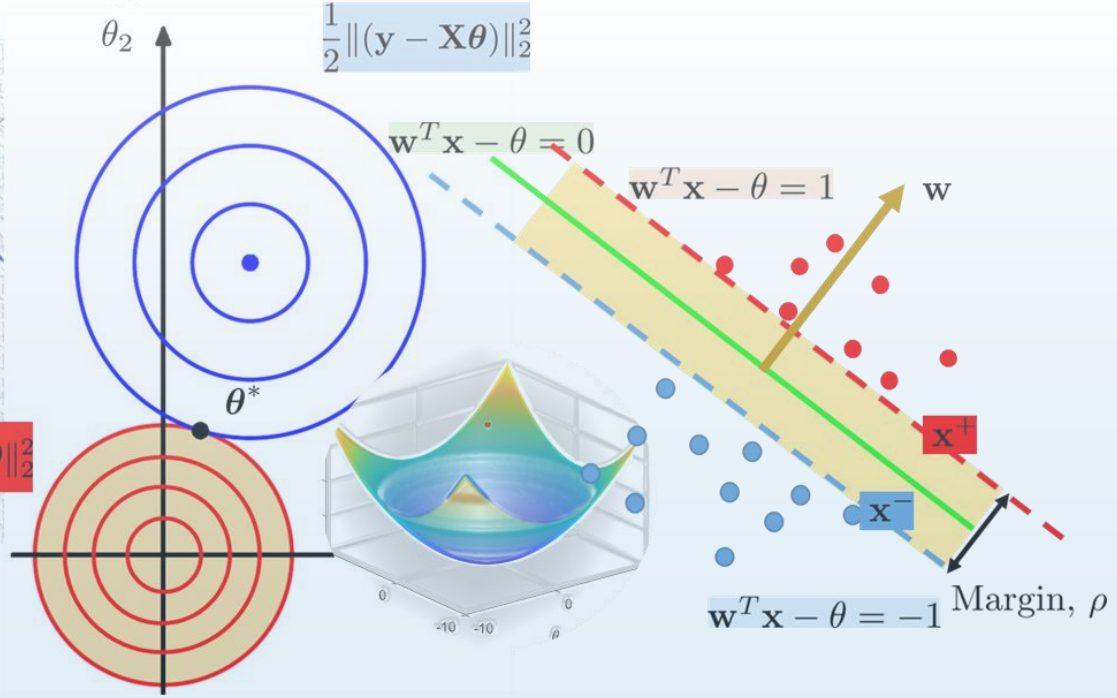




AI-501 Mathematics for AI

Overview

Zubair Khalid
School of Science and Engineering

LUMS
A Not-for-Profit University

Outline

- Course Overview, Logistics
- AI Overview
- Introduction to Machine Learning

About us!



Abdullah



Wazzah



Nausherwan



Sadaf



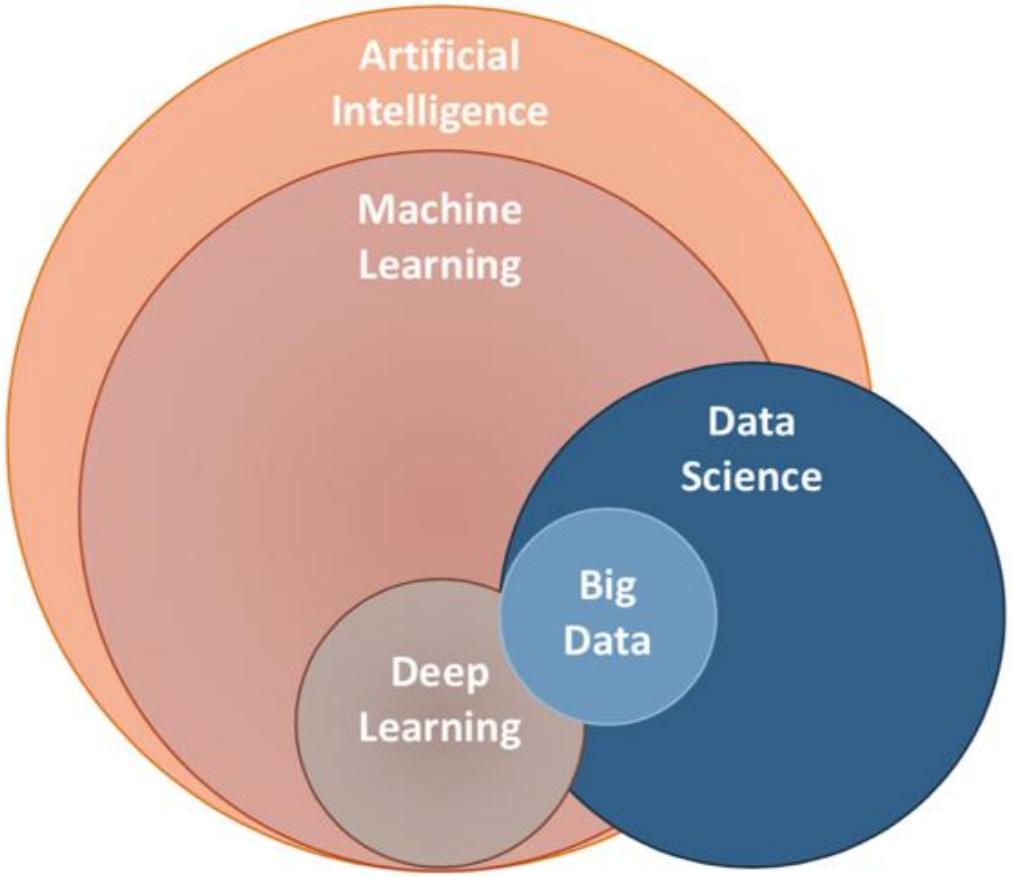
Zubair



Tayyab

LUMS
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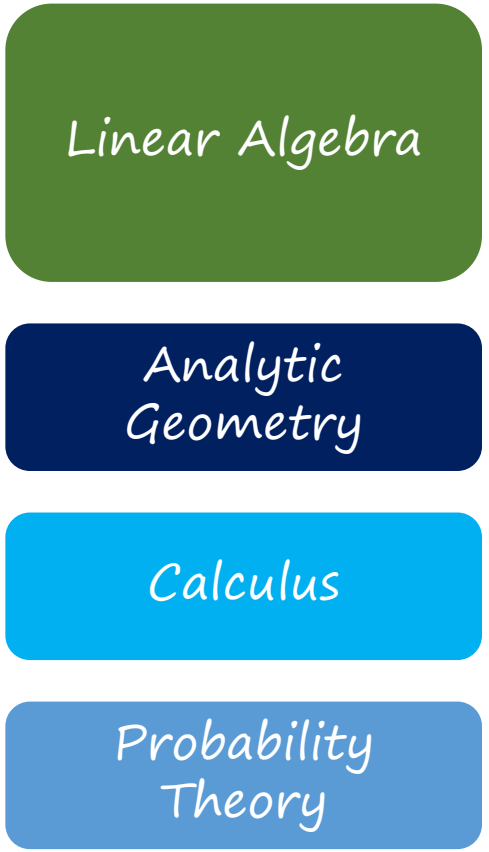
Overview



Is it possible to learn artificial intelligence without mathematical knowledge?

AI and Machine Learning

- Data
  - visualizations, operations, manipulations
- Analysis
  - insights, statistics, inference
- Model
  - representation of process, assumptions
- Learning
  - Optimize model with respect to data



Course Objectives

- **Foundational Knowledge:** Equip students with a robust understanding of linear algebra, calculus, probability theory, and optimization techniques, which are essential for ML and AI.
- **Practical Applications:** Demonstrate the application of mathematical concepts in real-world ML and AI scenarios, including data processing, model training, and performance evaluation.
- **Analytical Thinking:** Foster analytical thinking by encouraging students to critically assess different algorithms and their suitability for various tasks, considering both theoretical foundations and practical applications.

Is this course a right choice for you?

Graduate students

- Interesting in pursuing a **professional career** focused on the development of AI solutions
- Interested in pursuing AI, Deep Learning (DL) and Machine Learning (ML) in their **grad school (PhD)**
- Want to do fundamental research in the area of AI
- Wish to apply AI in their research work

Pre-requisites

- Encouraged to revise Linear Algebra and Probability concepts

Learning Interface

Communication and Interaction:



- Class:
  - Sunday, 9 am to 12 pm Slot
  - Break: 10:15 to 10:30
  - Quiz in class
  - Engagement in class

Course Page: LMS

Slack: Course-related questions or discussions. We will try to respond to the queries ASAP.
https://join.slack.com/t/ai-501/shared_invite/zt-478ysrlmd-MPpLco6TgMgmrPpQ6bHXuA

Office Hours: Posted on course page; distributed throughout the week. TAs during office hours will be available online as well.

Grading Distribution and Logistics

- CP: 5%
- Quizzes: 25% (N-X), X=2
- Mid-Exam: 30%
- Final Exam: 40%

Course Polices

- Missed Quiz Policy
  - No make-up for quiz
- Plagiarism will be strictly dealt with as per university policies (take it seriously).
- Zero Tolerance for Plagiarism and Cheating

Course Polices

Harassment Policy

Harassment of any kind is **unacceptable**, whether it be sexual harassment, online harassment, bullying, coercion, stalking, verbal or physical abuse of any kind. Harassment is a very broad term; it includes both direct and indirect behaviour, it may be physical or psychological in nature, it may be perpetrated online or offline, on campus and off campus. It may be one offense, or it may comprise of several incidents which together amount to sexual harassment. It may include overt requests for sexual favours but can also constitute verbal or written communication of a loaded nature. Further details of what may constitute harassment may be found in the LUMS Sexual Harassment Policy, which is available as part of the university code of conduct.

LUMS has a Sexual Harassment Policy and a Sexual Harassment Inquiry Committee (SHIC). Any member of the LUMS community can file a formal or informal complaint with the SHIC. If you are unsure about the process of filing a complaint, wish to discuss your options or have any questions, concerns, or complaints, please write to the Office of Accessibility and Inclusion (OAI, oai@lums.edu.pk) and SHIC (shic@lums.edu.pk) —both of them exist to help and support you and they will do their best to assist you in whatever way they can.

To file a complaint, please write to harassment@lums.edu.pk.

Course Polices

Help related to equity and Belonging at SSE

SSE's Council on Equity and Belonging is committed to devising ways to provide a safe, inclusive, and respectful learning, living, and working environment for its students, faculty, and staff.
For help related to any such issue, please feel free to write to any member of the school council for help or feedback.

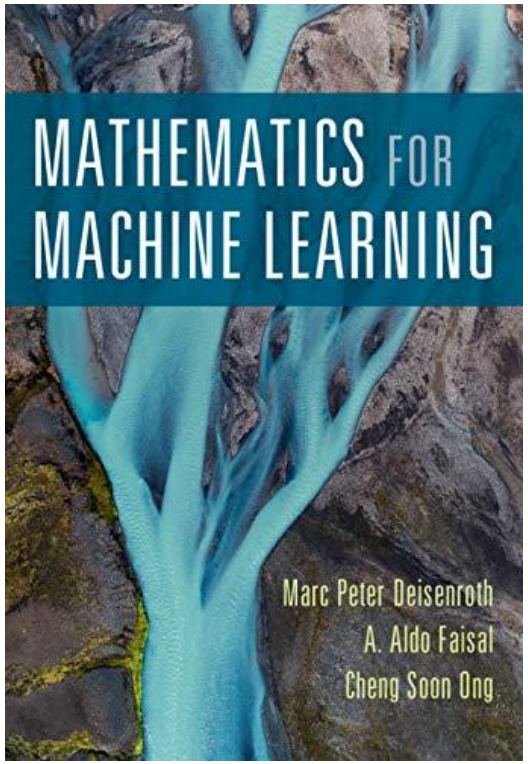
Mental Health Support at LUMS

-

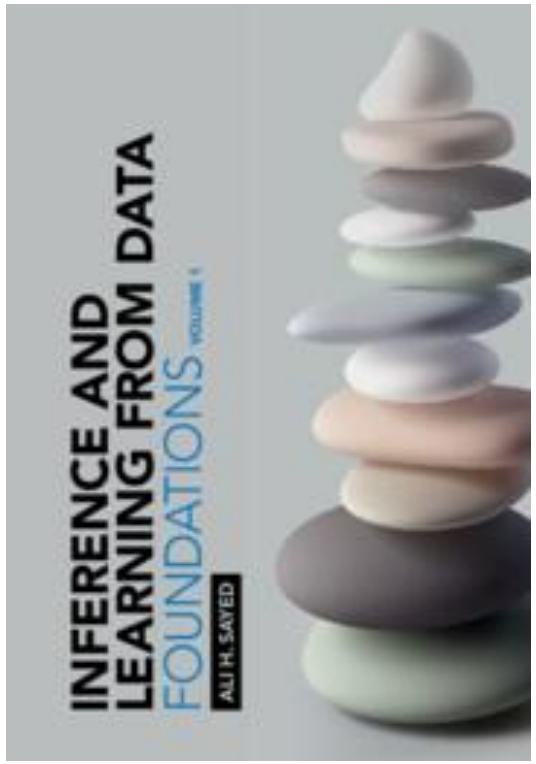
For matters relating to counselling, kindly email student.counselling@lums.edu.pk, or visit https://osa.lums.edu.pk/content/student-counselling-office for more information.

You are welcome to write to me or speak to me if you find that your mental health is impacting your ability to participate in the course. However, should you choose not to do so, please contact the Counselling Unit and speak to a counsellor or speak to the OSA team and ask them to write to me so that any necessary accommodations can be made.

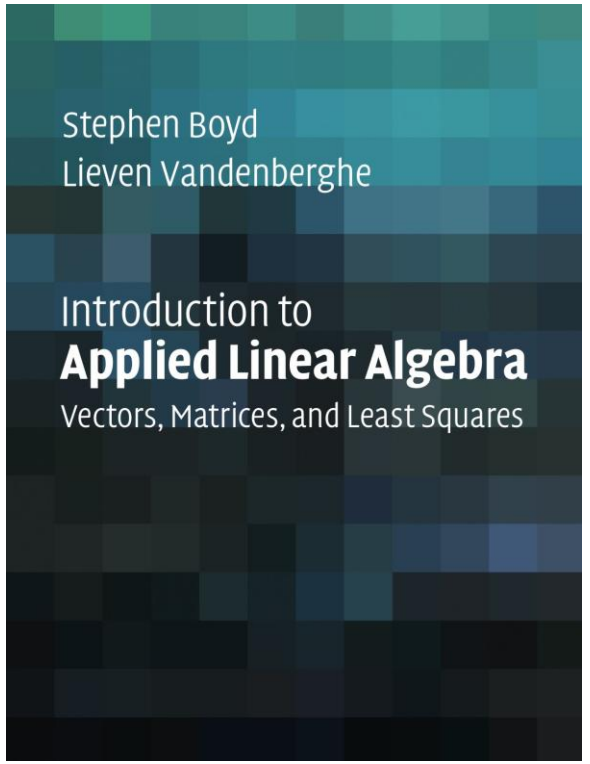
Suggested Reference Books



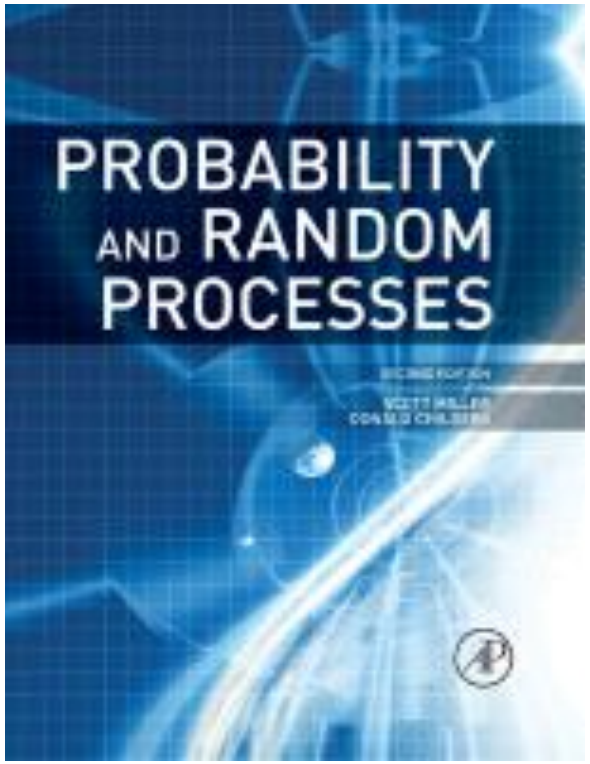
Primary



Reference



Reference



Reference

- Lecture Notes/Slides will be shared.

"As to methods, there may be a million and then some, but principles are few. The man who grasps principles can successfully select his own methods."

Ralph Waldo Emerson

Outline

- Course Overview, Logistics
- AI Overview
- Introduction to Machine Learning

AI Overview

What is Intelligence?

"Intelligence is the ability to **learn**, **understand**, and **apply** knowledge to adapt to new situations, reason, and solve problems."

- Human Intelligence
  - Problem-solving and reasoning
  - Learning from Experience
  - Adaptability
  - Emotional Intelligence

AI Overview

What is Artificial Intelligence (AI)?

"AI refers to the simulation of **human intelligence** in machines that are programmed to think and learn like humans."

- Narrow AI vs General AI
- SuperIntelligence



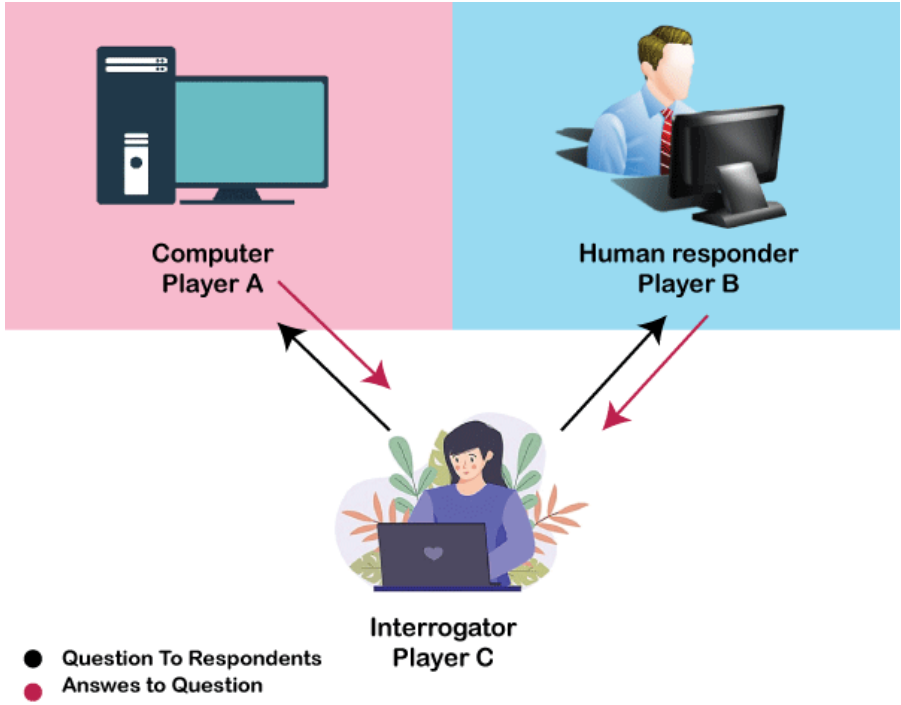
The term AI was coined in 1956 by John McCarthy at a conference.

AI Overview

History of AI – Turing Test

Alan Turing (1950) "Computing machinery and intelligence"

- **Can machines think or behave intelligently?**
- The Turing Test (imitation game) was designed as a way to judge the success or otherwise of an attempt to produce a thinking computer.
- Predicted that by 2000, a machine would have a 30% chance of fooling a person for 5 minutes
- What do you think?



AI Overview

History of AI – 1960s to 1990s

- Since the 1950s, much of the initial optimism surrounding AI has given way to a more realistic perspective.

- Key areas of focus include:
  - Machine learning
  - Multi-agent systems
  - Computer vision
  - Planning
  - Game Theory
  - Optimization, …

- 1997 (IBM's Deep Blue), 2012 (Deep Learning breakthrough).

Applications of AI

AI in Healthcare

- **AI-powered Diagnostics**

- AI systems can analyze medical images, such as X-rays, MRIs, and CT scans, to detect diseases like cancer, heart conditions, and more.

- These systems can identify patterns that might be missed by human eyes, resulting in earlier and more accurate diagnoses.

- **Example:** Google's DeepMind developed an AI system capable of diagnosing eye diseases with an accuracy that matches world-leading doctors.

Applications of AI

AI in Healthcare

- **Predictive Analytics**

- AI can analyze patient data and predict potential health risks, such as the likelihood of developing chronic diseases like diabetes or heart disease. This helps doctors create personalized treatment plans and intervene early.

- **Example:** IBM Watson Health analyzes vast datasets to predict disease progression and recommend personalized treatment options.

Applications of AI

AI in Healthcare

- **Robotic Surgery**

- AI-powered robotic systems assist in precision surgeries, allowing for less invasive procedures, faster recovery times, and reduced human error.

- **Example:** The da Vinci Surgical System uses AI to enhance the precision and control of surgeons during minimally invasive procedures.

Applications of AI

AI in Healthcare

- **Drug Discovery and Development**

- AI accelerates the drug discovery process by analyzing complex biological data to identify new drug candidates, significantly reducing the time and cost involved in developing new treatments.

- **Example:** AI was used by the pharmaceutical company Insilico Medicine to identify potential new compounds for treating fibrosis, which traditionally would have taken years.

Applications of AI

AI in Healthcare

- **Virtual Health Assistants**

- AI chatbots and virtual assistants provide personalized health advice, answer questions, and help patients manage chronic conditions by monitoring symptoms and offering medication reminders.

- **Example:** AI-driven platform allows patients to receive medical advice through their smartphones based on the symptoms they input.

Applications of AI

AI in Finance

- Fraud Detection
- Algorithmic Trading
- Risk Management
- Personalized Banking
- Credit Scoring and Loan Decisions
- Financial Forecasting

Applications of AI

AI in Education

- Personalized Learning Systems
- Virtual Tutors and Chatbots
- AI in Assessments and Grading
- AI in Special Education

Applications of AI

AI in Autonomous Vehicles

- Deep Learning for Object Detection
- Decision Making and Planning (Reinforcement Learning)
- Simultaneous Localization and Mapping (SLAM)
- Path Planning and Navigation
- To name a few…

Applications of AI

AI in Retail

- Personalized product recommendations
- AI-powered chatbots and virtual assistants
- Inventory management and demand forecasting
- Dynamic pricing
- Visual search and image recognition
- Customer sentiment analysis

Applications of AI

AI in Agriculture

- Precision farming
- Crop monitoring and disease detection
- Soil health monitoring and analysis
- Automated machinery and robotics
- Predictive analytics and yield forecasting

Applications of AI

AI in NLP

- Virtual assistants (e.g., Siri, Alexa)
- Language translation
- Text generation and summarization
- Speech recognition and transcription
- Sentiment analysis
- Chatbots for customer support

Applications of AI

AI in Manufacturing

- Predictive maintenance
- Quality control and defect detection
- Supply chain optimization
- Autonomous robots and cobots (collaborative robots)
- Inventory management and optimization
- Production scheduling and process optimization

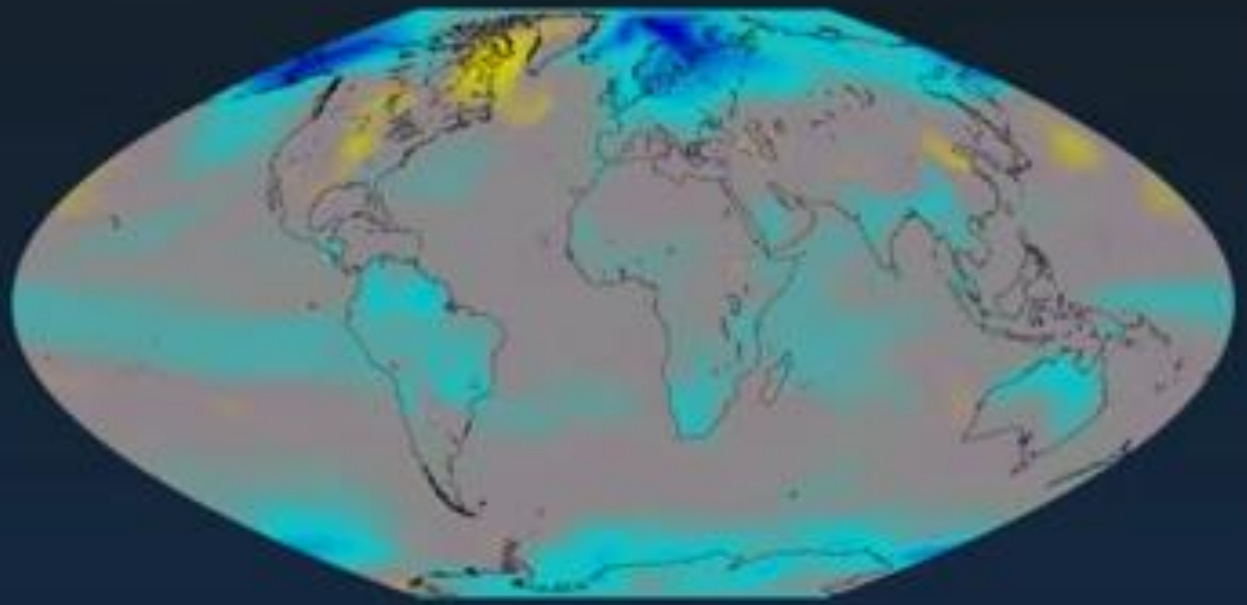
Applications of AI

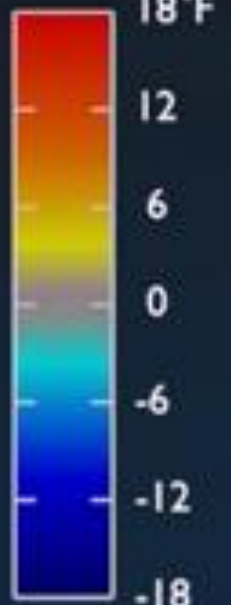
AI in Urban Development

- Traffic management and optimization
- Smart energy grids and resource management
- Road safety and surveillance
- Waste management optimization
- Smart infrastructure and building management
- Environmental monitoring

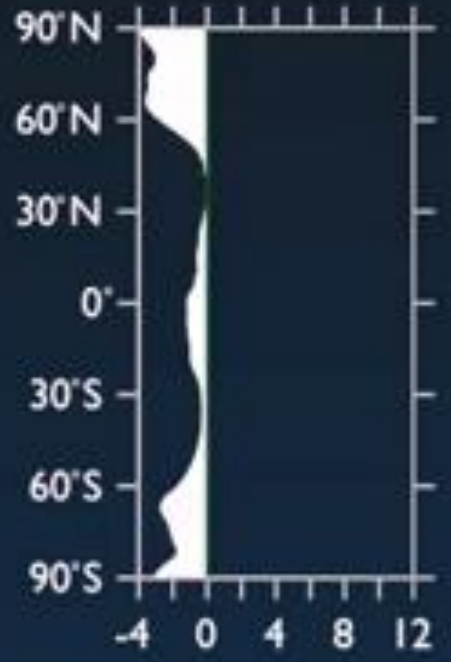
CITY Overview

Centre for Urban Informatics, Technology, and Policy (CITY)

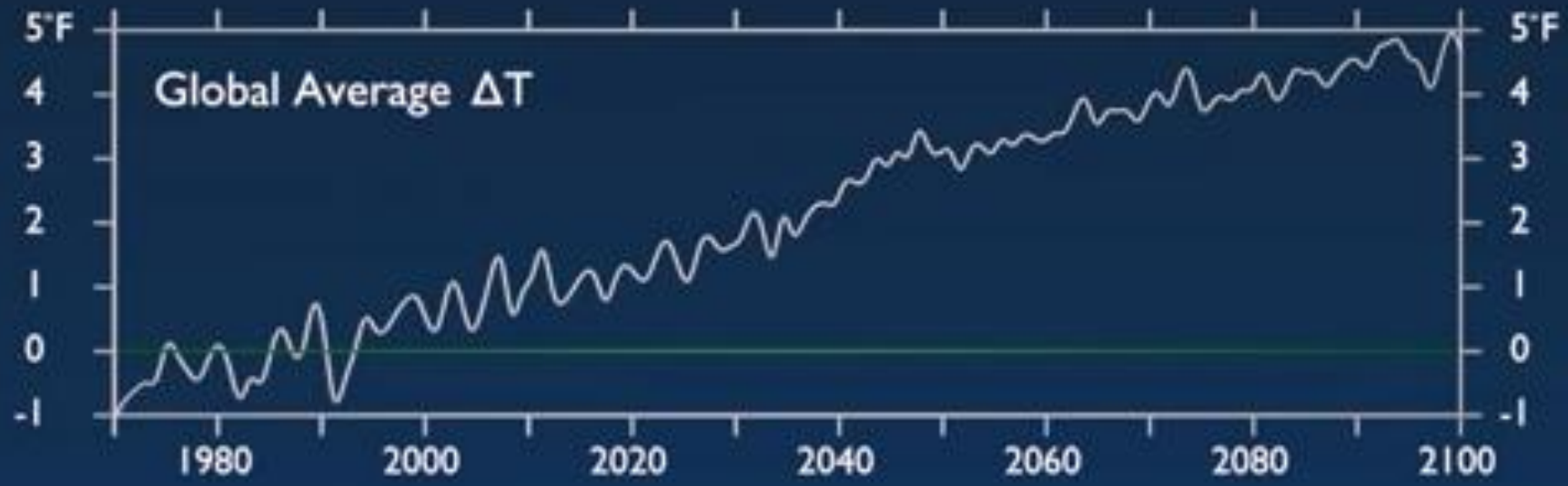




ΔT Anomalies Relative to 1971-2000 Mean



Zonal Average (°F)



1971
(Model Year)

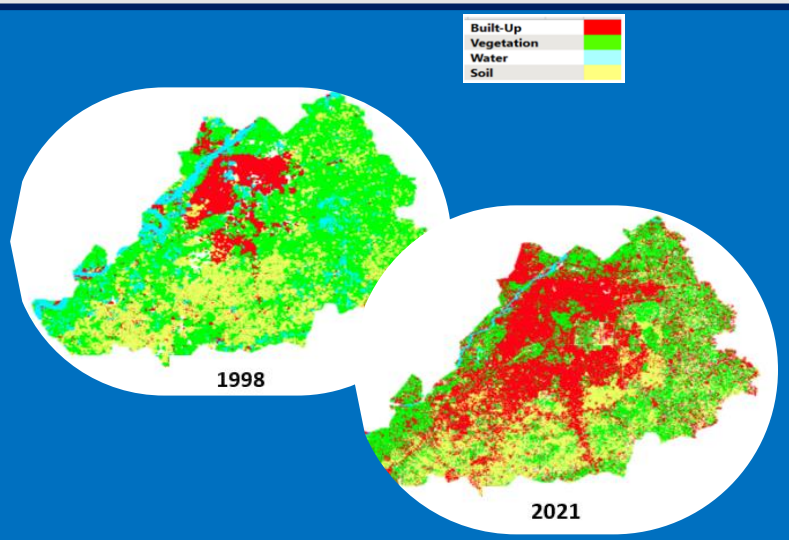
GFDL CM 2.1
Climate Model


Lahore
New Delhi
10 NOV. 2025

The Grand Challenge

Rapid urbanization, **resource scarcity** and **climate change** are testing the resilience of cities

Growth



Lahore's Built-up Area. 4.5X (1998 – 2022)

Mobility

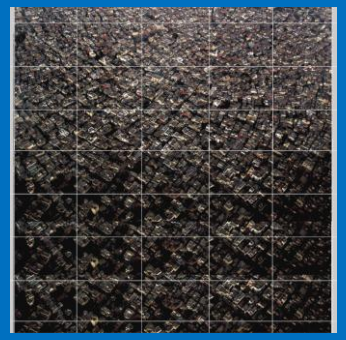


4x (2010-2024)

3x (2010-2024)

Environment and Health



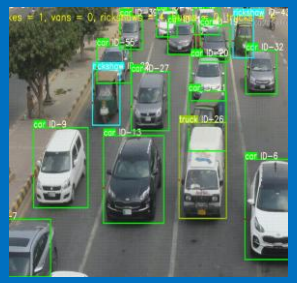


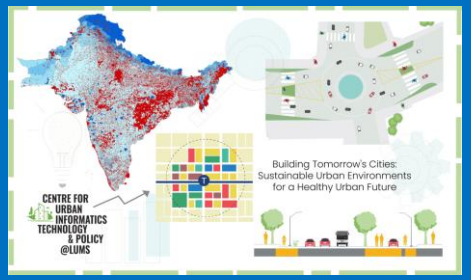
1 Research

Climate
Mobility
Space-time Resource Optimization

Solutions 2

Machine Vision & AI
Intelligent Traffic Insights
Data-Driven Policies





3 Outreach

Engagement with Government
International Collaborations, Grants
Seminars & Workshops

Academy 4

Workshops
Trainings
Urban Dictionary



CITY at LUMS

A center focusing on **climate-resilient urban solutions** through **technology, data, and policy integration**

AI, Optimization, Modeling, Policy
Deep Tech

↓

Data Generation and Analysis

↓

Evidence-Based Decision Making

↓

Sustainable Communities



Inclusivity

Equity

Urban Mobility: Road Network

Data Layers

Google Earth

Google Maps Platform

Open Street Maps

Manual Labeling

Machine Learning and Image Processing

Road Network Lahore



Objective:
Creating a dense road network with associated attributes like speed, distance, lanes, etc.

Road Network

Urban Mobility: Road Network

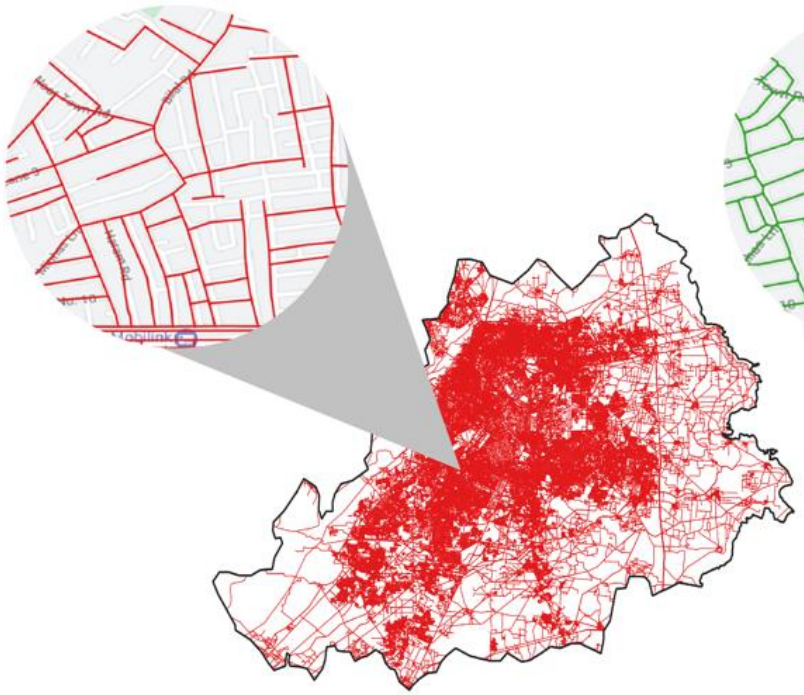


Analysis Tools:

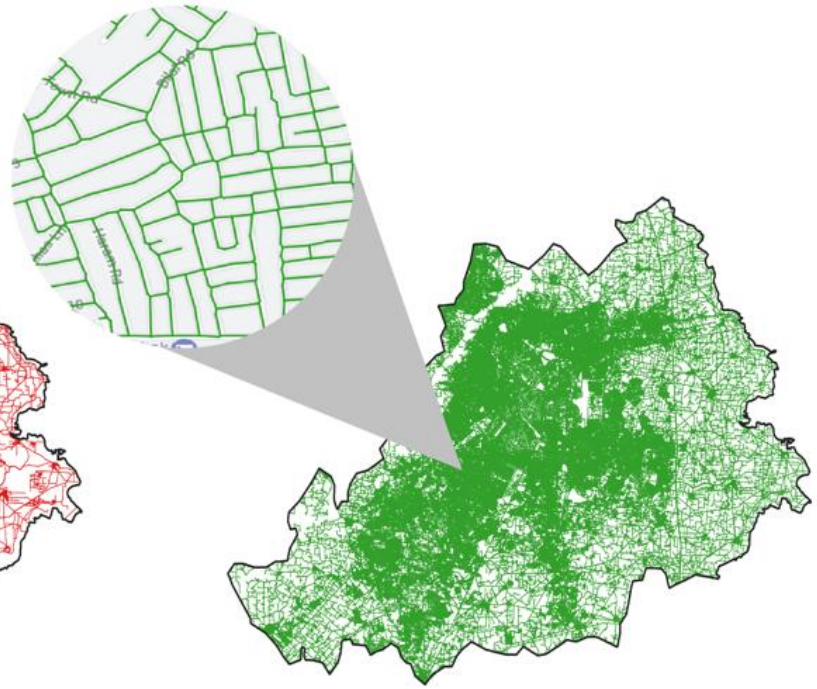
QGIS, Python, OpenCV, and SciPy

Road Network

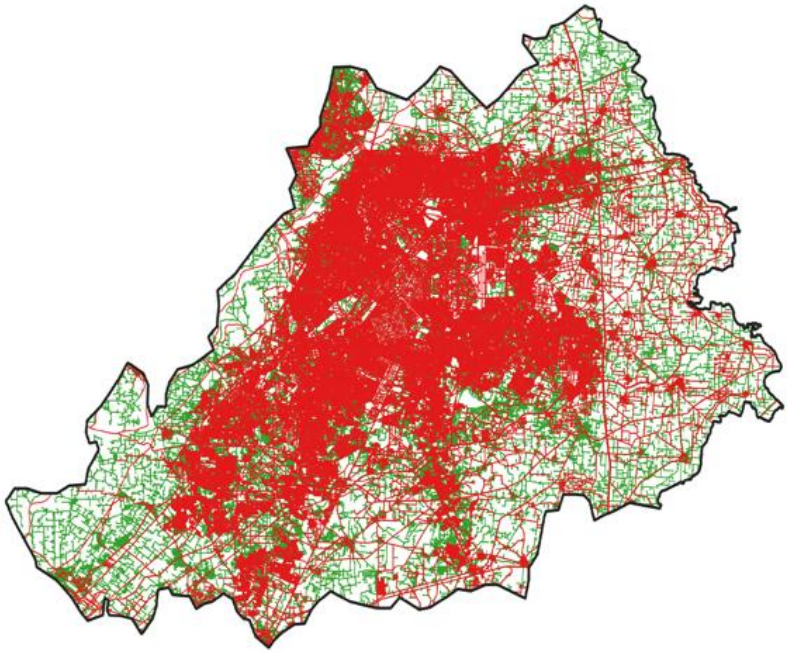
Urban Mobility: Road Network



OSM Road Network



Our Road Network



Overlaid

Urban Mobility: Traffic Informatics

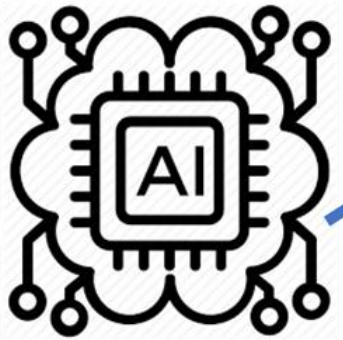
Road monitoring using camera



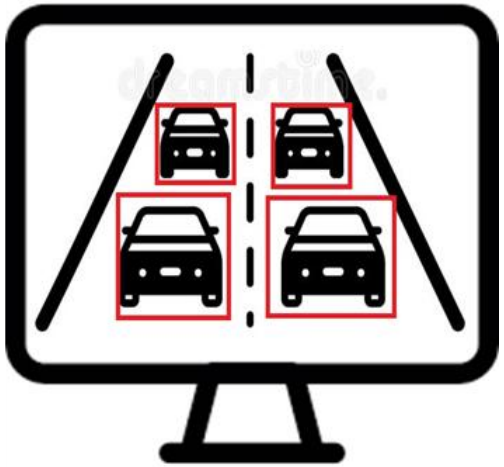
Database/Storage



AI model



Dashboard



AI based system for real time traffic flow density

Traffic Flows Density

Urban Mobility: Traffic Informatics

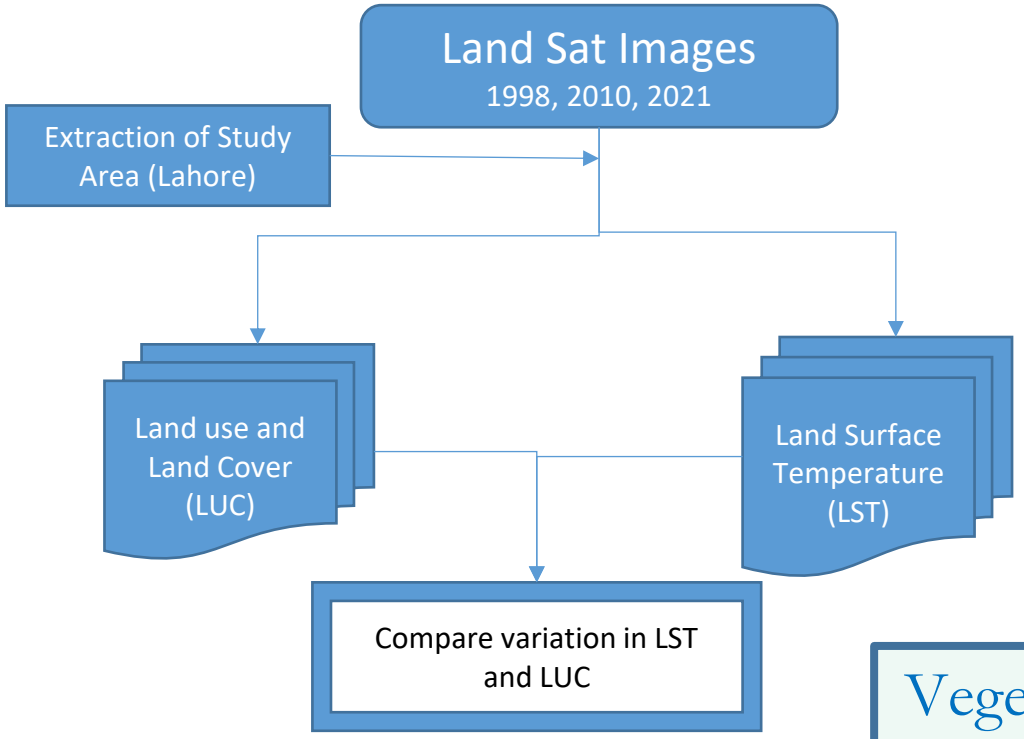


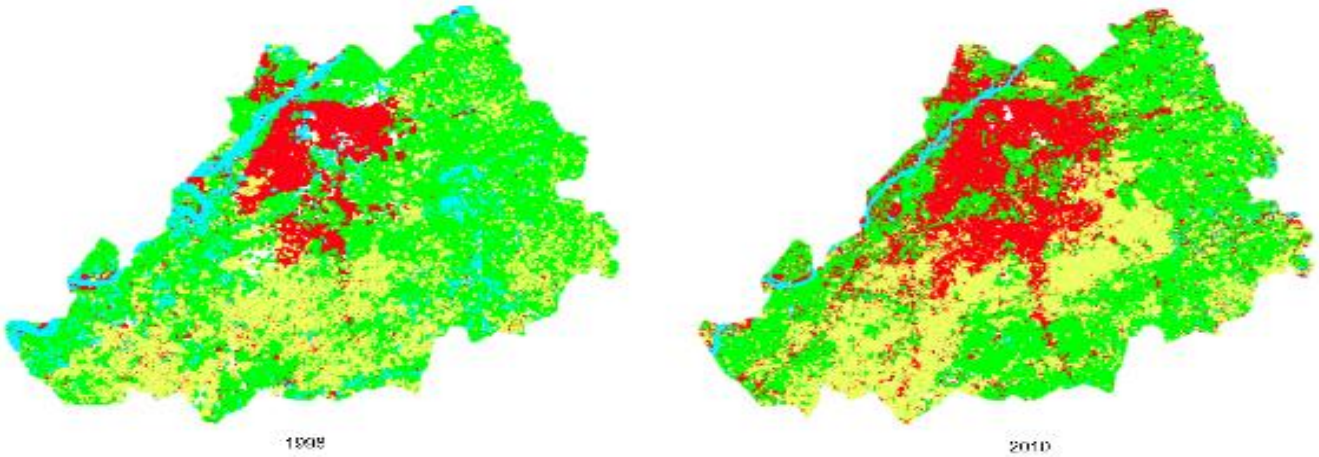
Prototype Sample

Traffic Flows Density

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Urban Sprawl: Land Surface Temperature





Vegetation and waterbodies declined; Built-up use grew significantly

An average increase of 0.47 Celsius per year between 1998 and 2021

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Data-driven Urban Forestation: Tree Cover Estimation

Problem: **Estimate green spaces and tree cover in urban areas**

Why?

Green spaces – significant contributors to ecosystem services: air purification and carbon sequestration

Challenges

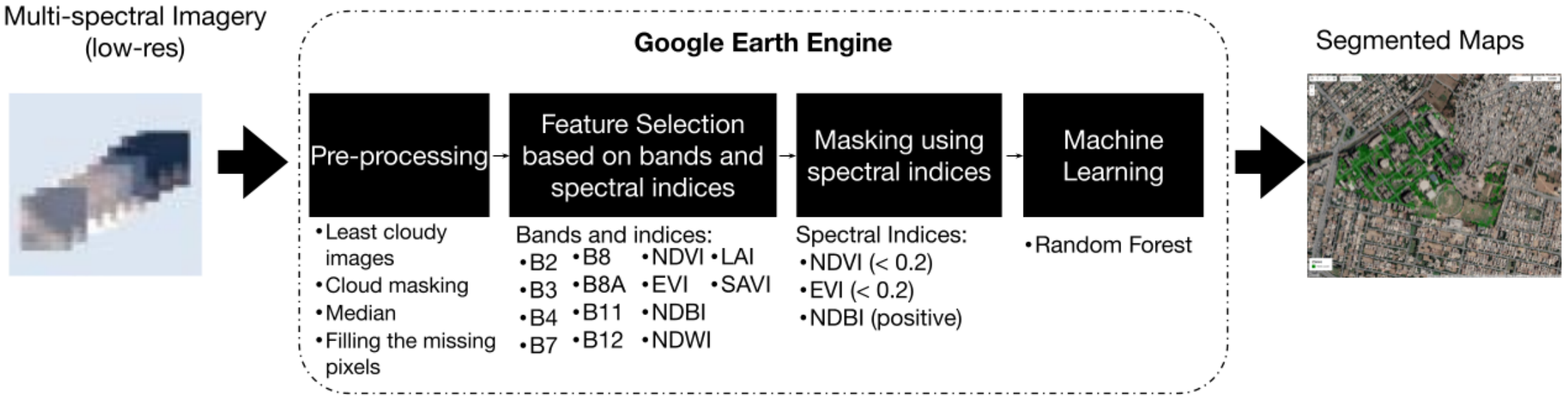
- Bottleneck: absence of spatially explicit data
- Huge variations in satellite imagery

Existing Methodologies

- Current methodologies focus on forest mapping
- A gap exists in urban tree cover estimation in developing countries with low resolution imagery

Data-driven Urban Forestation: Tree Cover Estimation

Tree Cover Estimation using Machine Learning and Feature Selection on Sentinel-2 Data



U. Nazir, M. Uppal, M. Tahir, and Z. Khalid, "Feature Selection on Sentinel-2 Multi-Spectral Imagery for Efficient Tree Cover Estimation," 2023 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2023), Pasadena, California

Data-driven Urban Forestation: Tree Cover Estimation



LUMS Ground Truth



Proposed



ESA WorldCover

Model	Pred. area (acres)	Masking	Spectral indices	Pixel-wise Test Accuracy (%)	Kappa Score
RF-spectral-bands	29.5	No	No	0.93	0.81
RF-spectral-indices	28	No	Yes	0.95	0.88
Proposed	25	Yes	Yes	**0.99**	**0.92**
ESA WorldCover Product [5]	16	-	-	0.74	-
DeepLabv3 [6]	28	No	No	0.80	-

CITY in action: four snapshots

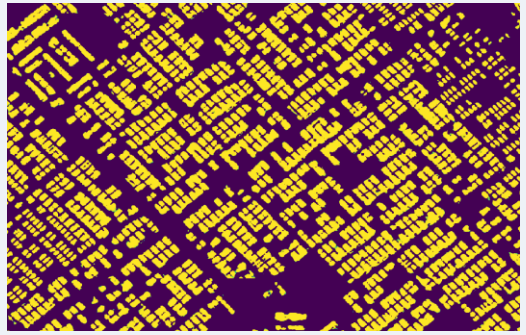
Mobility informatics

Real-time traffic density from intersection cameras.



Urban sprawl

Land use, building footprints, *Global Building Atlas.*



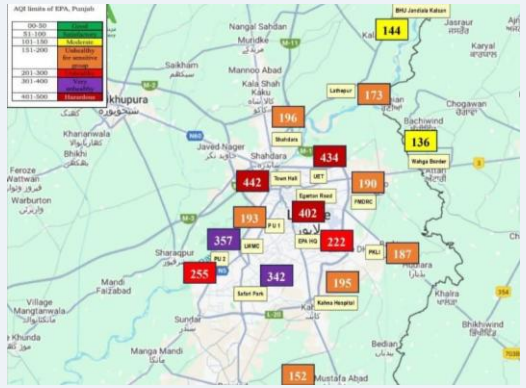
Emergency response

Accident hot-spots; optimised ambulance allocation.



Air quality

Dense PM2.5 sensing; smog-season forecasting.



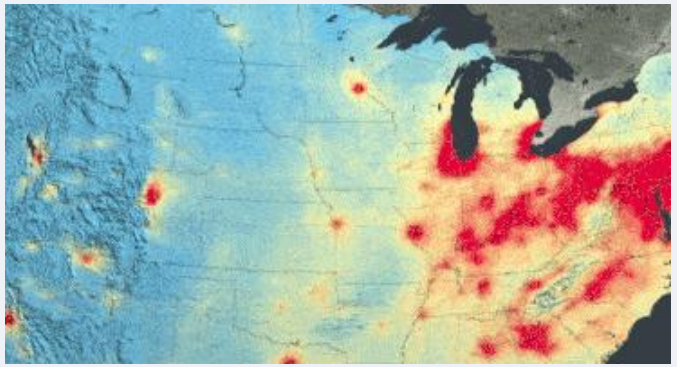
Every project turns a spatiotemporal signal into a decision.

The complexity of the physical world

Air Quality

PM2.5 sensor networks

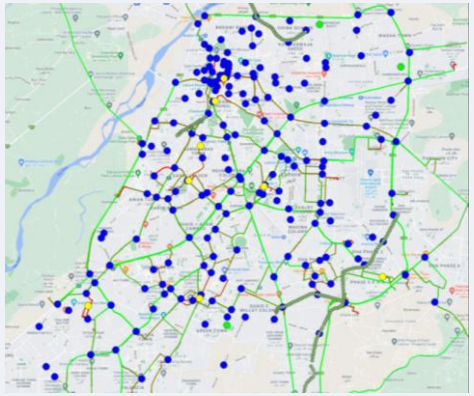
$y(s,t)$ irregular, non-stationary



Urban Mobility

Ride-hailing / traffic

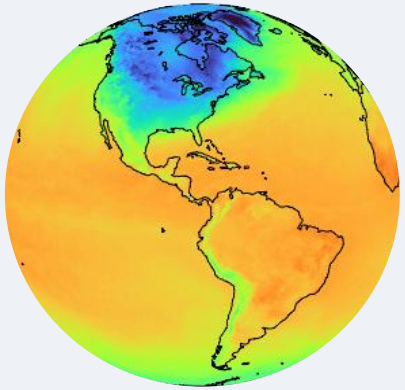
$X_t^{w,h}$ grid + graph



Climate

Global 2 m temperature

$y_t(\theta,\phi)$ on the sphere



Common ground: complex, non-stationary, spectrally entangled spatiotemporal signals.

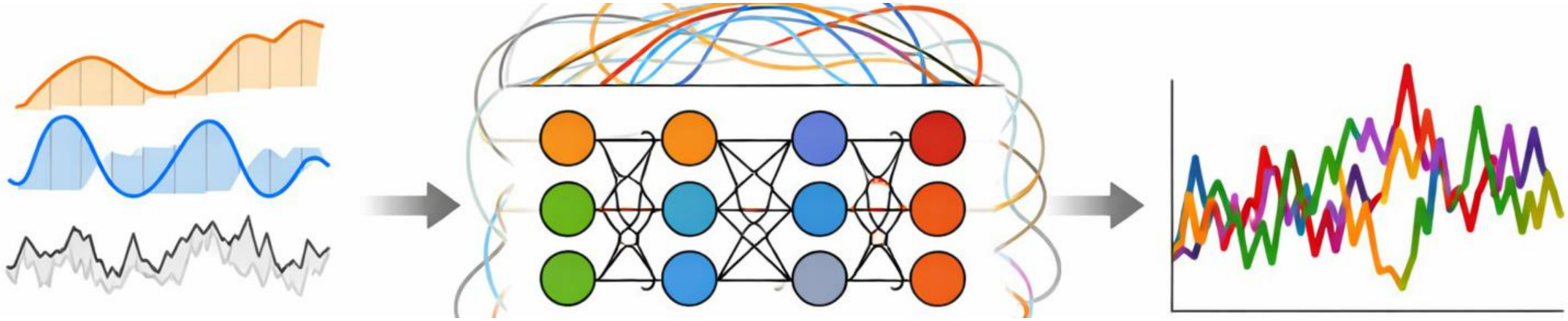
Different sensors, different geometry, same signal-processing question.

Why does deep learning struggle here?

Mixed Signals
(Non-stationary data)

Black Box
(Deep Learning)

Output
(Noisy, Chaotic)



- **Treats spatiotemporal signals as monolithic:** Ignores the frequency / mode structure of the data.
- **Spectral entanglement:** rush-hour harmonics + weather bursts + event spikes fold into one series.
- **Limited interpretability:** hard to attribute predictions to physical mechanisms.

Math-informed AI: decompose first, then learn on the pieces.

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- Course Overview, Logistics
- AI Overview
- Introduction to Machine Learning

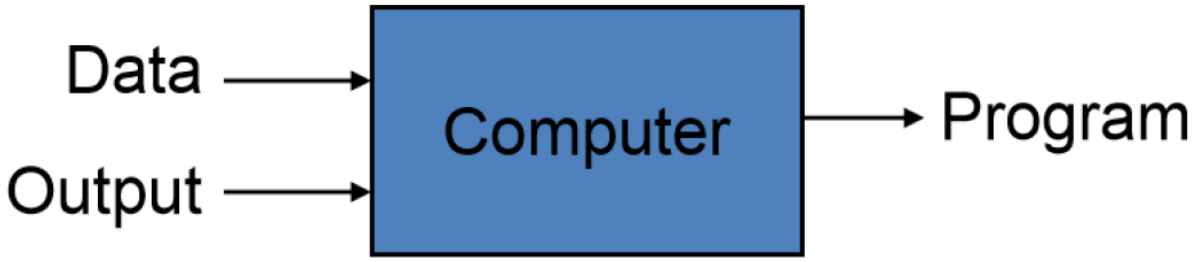
Machine Learning Overview

What is Machine Learning?

- Automating the process of automation
- Getting computers to program themselves



Traditional Programming

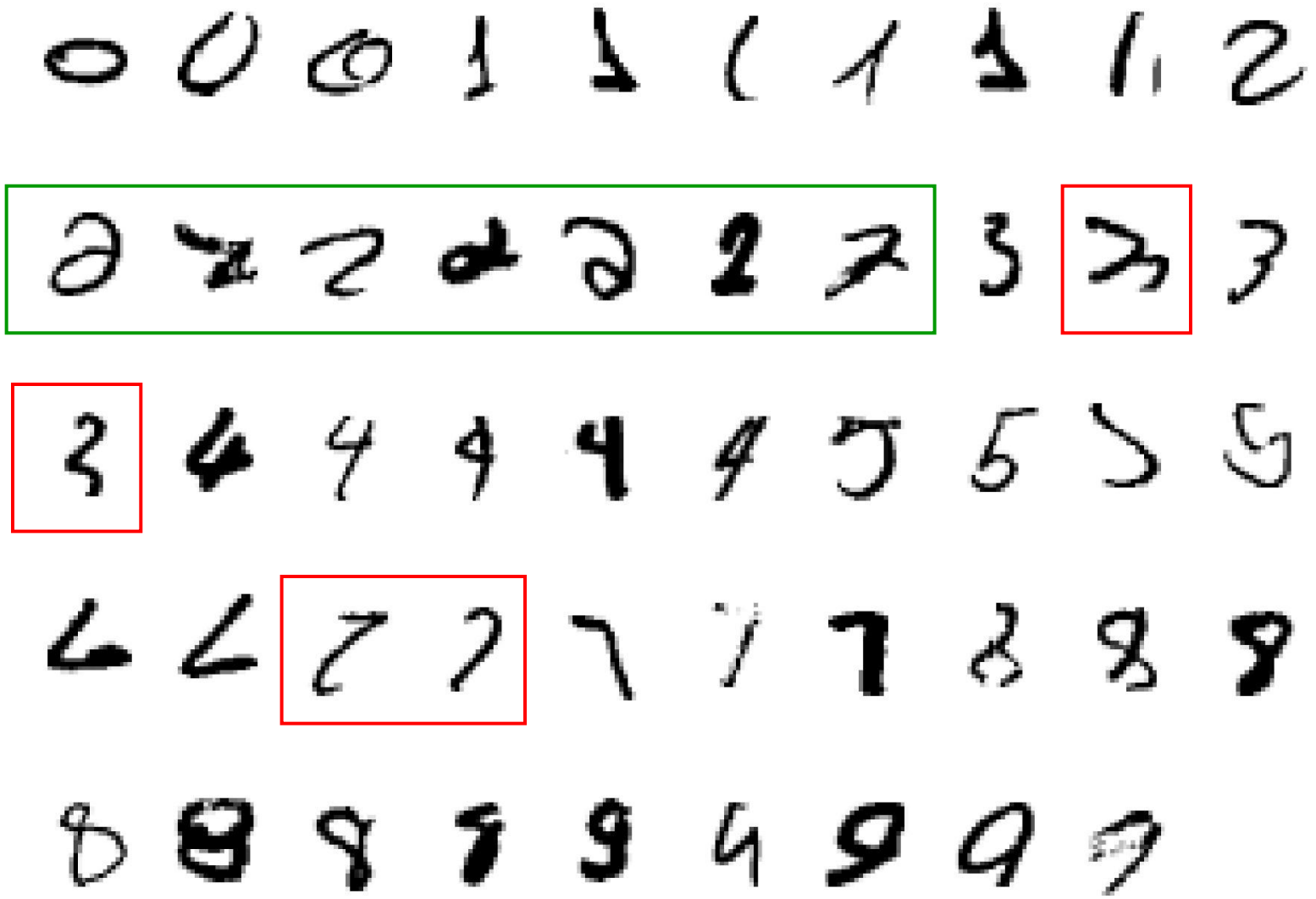


Machine Learning

Given examples (training data), make a machine learn system behavior or discover patterns

Machine Learning Overview

Classical Example: Recognize hand-written 2!

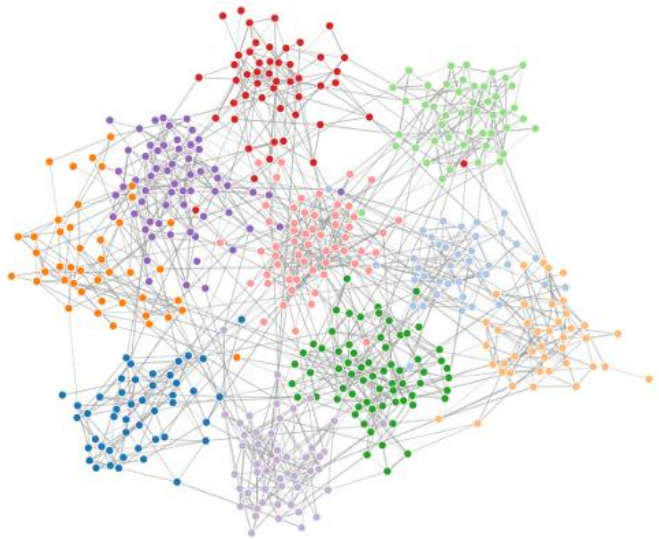


Machine Learning: Overview

What is Machine Learning?

Given examples (training data), make a machine learn system behavior or discover patterns

Data



Algorithm

Model

$$f(\mathbf{x})$$

Given to us

We need to design it

Final output which enables us to make predictions

Machine Learning: Overview

Algorithms vs Model

- Linear regression algorithm produces a model, that is, a vector of values of the coefficients of the model.

- Decision tree algorithm produces a model comprised of a tree of if-then statements with specific values.

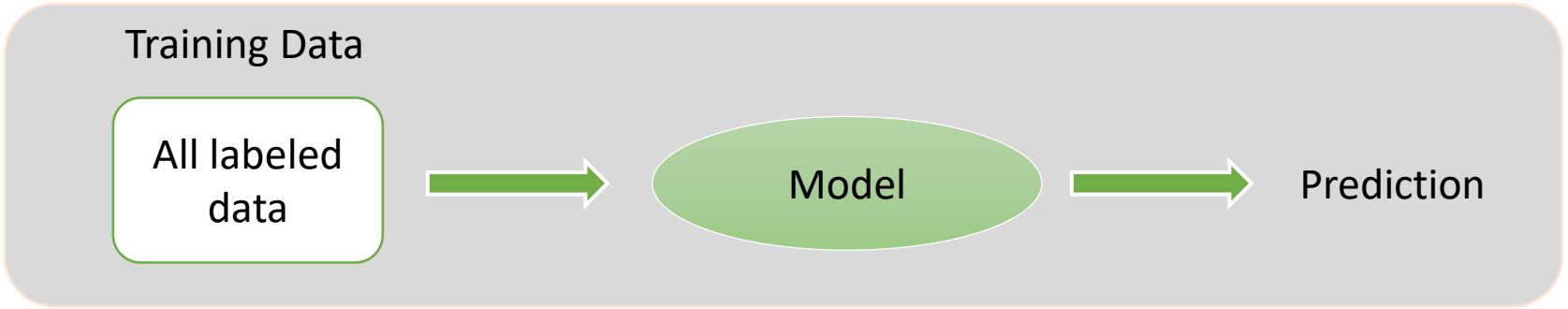
- Neural network along with backpropagation + gradient descent: produces a model comprised of a trained (weights assigned) neural network.

Machine Learning: Overview

Nature of ML Problems

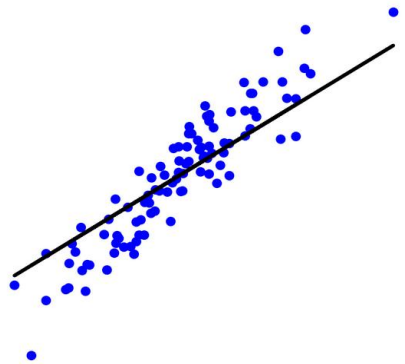
1. Supervised Learning

The learning algorithm would receive a set of inputs along with the corresponding correct outputs to train a model



Supervised Learning

Regression



Regression: Quantitative Prediction on a continuous scale

Examples: Prediction of

- Age of a person from his/her photo
- Price of 10 Marla, 5-bedroom house in 2050
- USD/PKR exchange rate after one year
- Efficacy of vaccine or medicine
- Average temperature/Rainfall during monsoon
- Cumulative score in AI501 course
- Probability of decrease in the electricity prices in Pakistan
- No. of steps per day

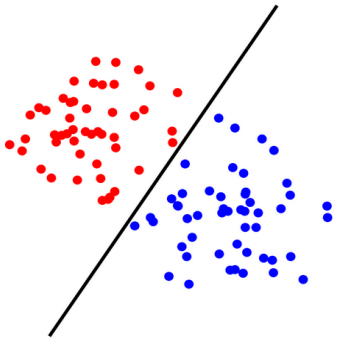
What do all these problems have in common?

Continuous outputs

Predicting continuous outputs is called regression

Supervised Learning

Classification



Classification: Given a data sample, predict its class **(discrete)**

Examples: Prediction of

- Gender of a person using his/her photo or hand-writing style
- Spam filtering
- Object or face detection in a photo
- Temperature/Rainfall normal or abnormal during monsoon
- Letter grade in AI501 course
- Decrease expected in electricity prices in Pakistan next year
- More than 10000 Steps taken today

What do all these problems have in common?

Discrete outputs: Categorical

Yes/No (Binary Classification)

Multi-class classification: multiple classes

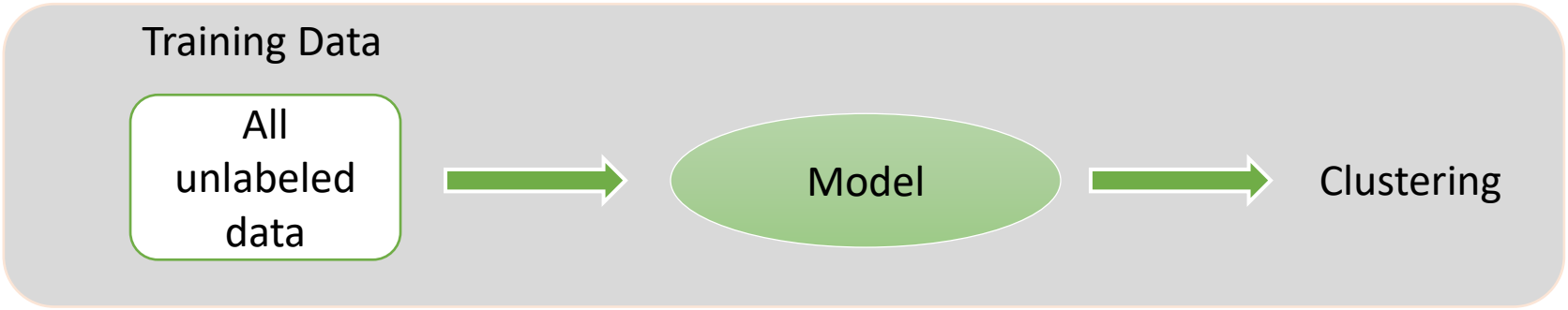
Predicting a categorical output is called classification

Machine Learning: Overview

Nature of ML Problems

2. Unsupervised Learning

The learning algorithm would receive unlabeled raw data to train a model and to find patterns in the data



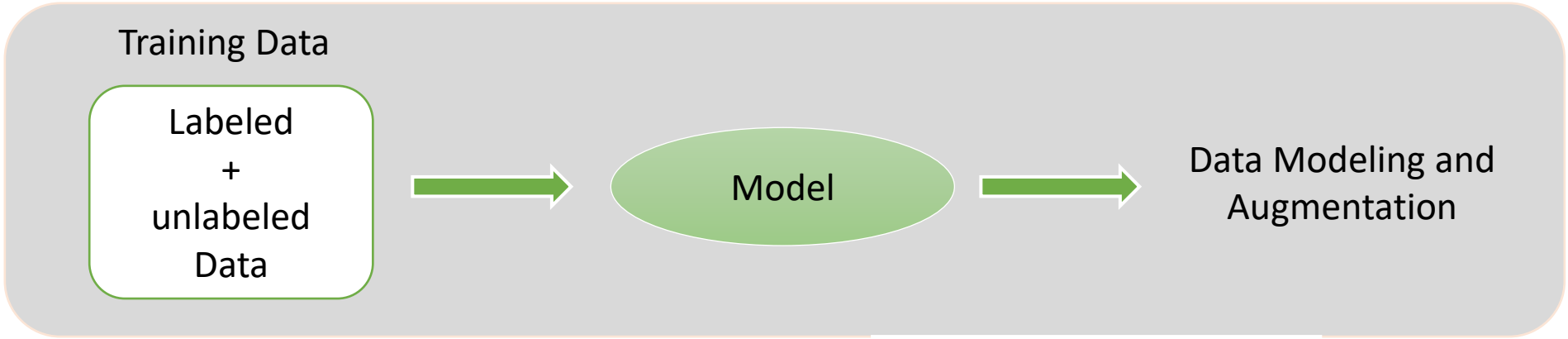


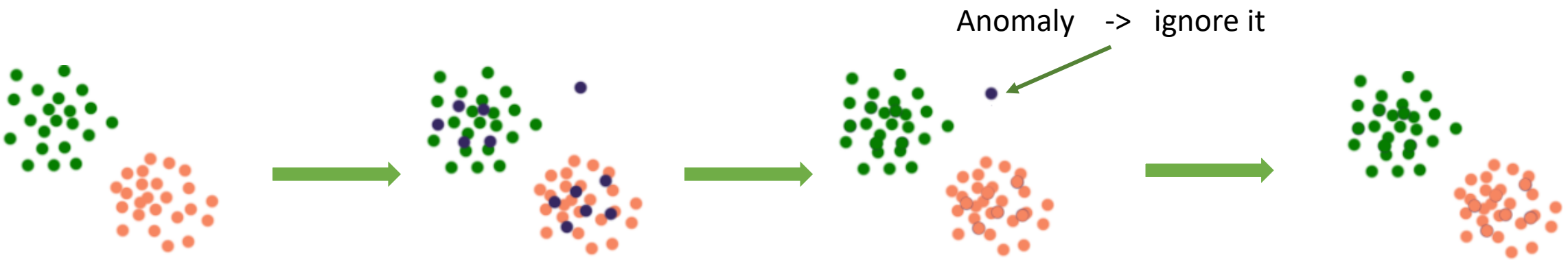
Machine Learning: Overview

Nature of ML Problems

3. Semi-supervised Learning

- *The learning algorithm receives labeled and unlabeled raw data to train a model*
- *Main objective is to efficiently accommodate the unlabeled data*





Use labeled data to build a model

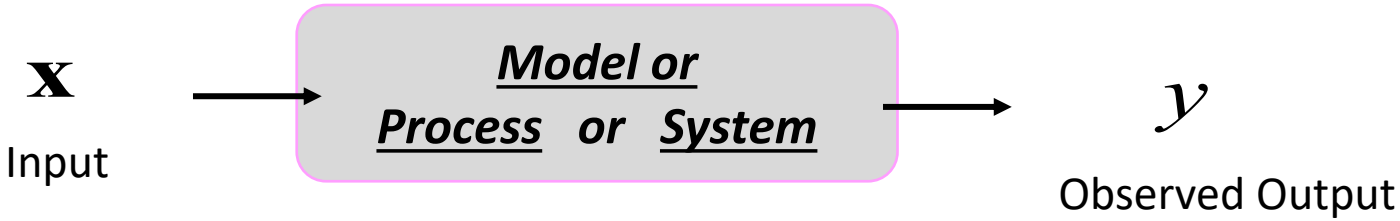
Place unlabeled data with model

Use the model to label the unlabeled data

Fit the model again with the combined data

Machine Learning: Overview

Training Data Collection

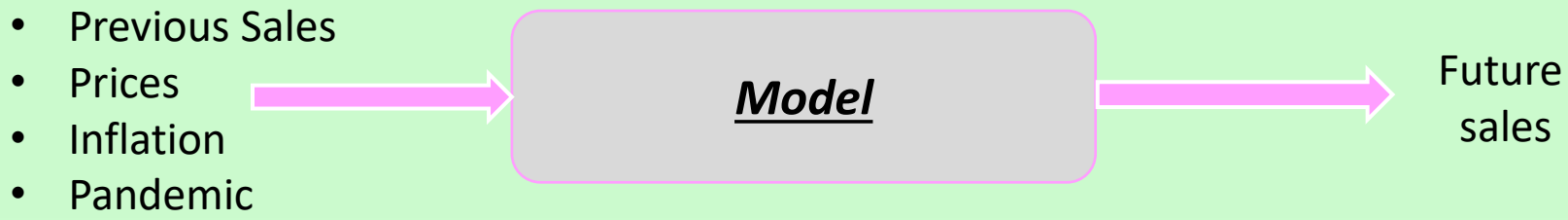


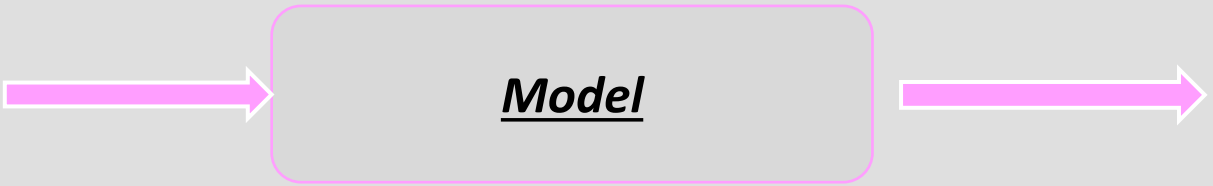
PROCESS or ***SYSTEM*** : Underlying physical or logical phenomenon which maps our input data to our observed output

Collect the training data by observing our unknown **PROCESS** or **SYSTEM**

Machine Learning: Overview

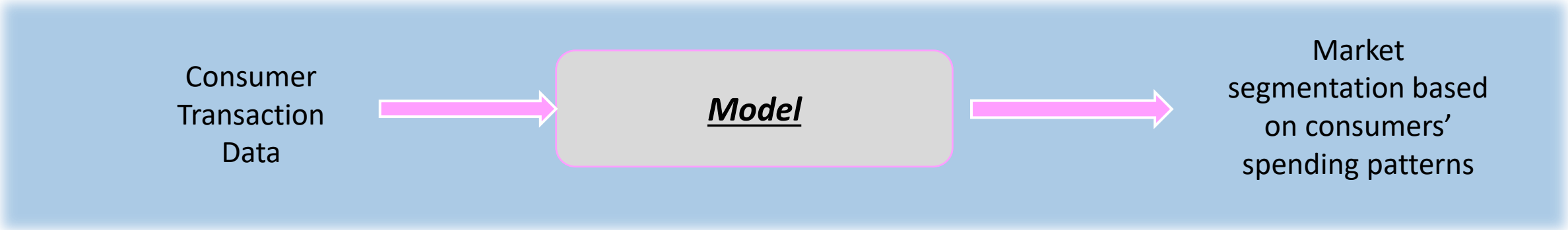
Example Systems

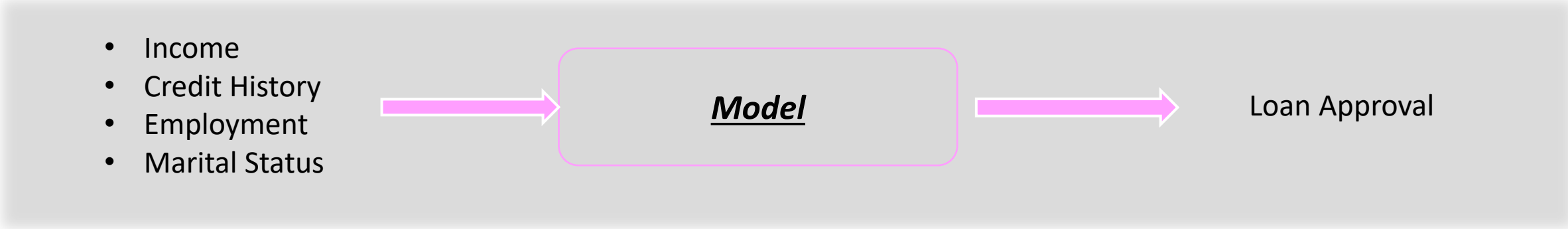




Machine Learning: Overview

Example Systems

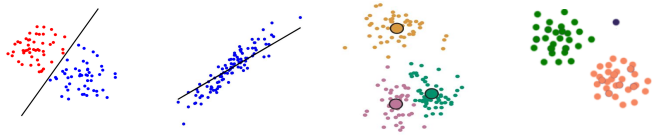




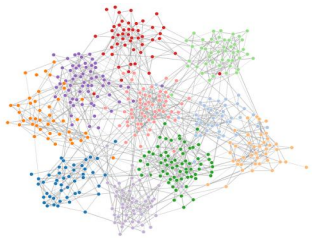
Machine Learning: Overview

Typical Flow

Problem Nature Identification



Training Data Collection



Model Training

$$f(\mathbf{x})$$

Model Validation



Model Deployment



Supervised Learning Setup

Nomenclature

In these regression or classification problems, we have

- Inputs – referred to as Features
- Output – referred to as Label
- Training data – (input, output) for which the output is known and is used for training a model by ML algorithm
- A Loss, an objective or a cost function – determines how well a trained model approximates the training data
- Test data – (input, output) for which the output is known and is used for the evaluation of the performance of the trained model

Supervised Learning Setup

Nomenclature - Example

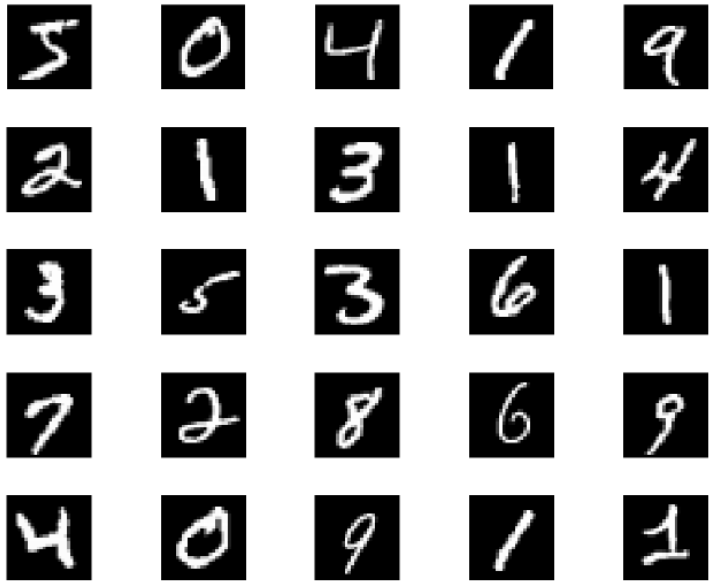
Predict Stock Index Price

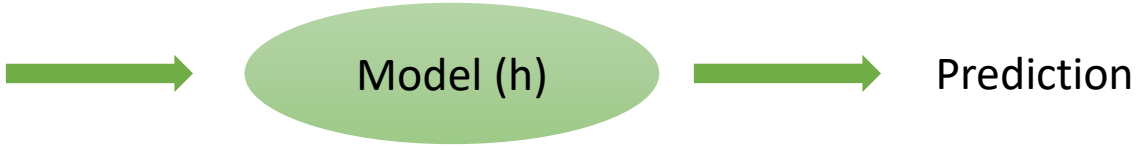
- *Features (Input)*
- *Labels (Output)*
- *Training data*

Interest_Rate	Unemployment_Rate	Stock_Index_Price
2.75	5.3	1464
2.5	5.3	1394
2.5	5.3	1357
2.5	5.3	1293
2.5	5.4	1256
2.5	5.6	1254
2.5	5.5	1234
2.25	5.5	1195
2.25	5.5	1159
2.25	5.6	1167
2	5.7	1130
2	5.9	1075
2	6	1047
1.75	5.9	965
1.75	5.8	943
1.75	6.1	958
1.75	6.2	971
1.75	6.1	949
1.75	6.1	884
1.75	6.1	866
1.75	5.9	876
1.75	6.2	?
1.75	6.2	?
1.75	6.1	?

Supervised Learning Setup

Example





MNIST Data:

- Each sample 28x28 pixel image
- 60,000 training data
- 10,000 testing data